

Intercropping with perennial herbs and flowering plants

Spatial scale: vineyard

About this sustainable management strategy

Cover crops are plants that are intentionally grown between the vine rows. They are meant to enhance soil properties without the need to use herbicides. Management involves various application methods, such as planting in all or alternating rows of the most appropriate native or locally adapted plant species. There is a wide selection of possible plant species that can be used as cover crops. Common species include annuals like barley and oats, legumes such as vetch and clover, and brassicas like daikon radish. The selection of species to seed and the timing depend on the desired effect and on the season. For example, sowing in autumn is meant to provide a green manure to a vineyard over the winter period, while mowing cover crops in summer creates a mulch layer that reduces water evaporation from the soil during warm summer periods. After the growing period, the crops can then be ploughed under and incorporated into the soil as green manure to improve soil fertility.

Key ecosystem services

- Soil health and fertility (*improved soil structure, nutrient cycling and organic matter content*)
- Erosion control (*slope stabilisation*)
- Carbon sequestration
- Water retention (*reduced surface runoff*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Biodiversity enhancement
- Natural pest control (*support for beneficial insects and natural enemies*)
- Climate regulation (*local microclimate regulation, reduced temperature extremes*)
- Cultural heritage (*preservation of traditional knowledge and historical land-use practices*)
- Landscape aesthetics (*distinct landscape character*)

Considerations for successful implementation

Resource interactions with vines

Resource interactions between cover crops and grapevines should be considered when designing and managing intercropping systems, particularly in young vineyards or water-limited environments. The choice of species, sowing density, and management regime largely determines whether ecological benefits are achieved without adversely affecting vine performance.

Management and labour requirements

The establishment and maintenance of cover crops require planning and regular management, including the timing of sowing, mowing, and incorporation practices. Successful implementation depends on integrating these activities into existing vineyard operations and adapting them to local conditions.

Site-specific performance

Cover crop performance varies according to climate, soil conditions, species selection, and seed quality. Tailoring mixtures and management practices to local conditions is often necessary to ensure reliable establishment, effective ground cover, and long-term ecosystem benefits.